

**SAFETY DATA SHEET**

Creation Date 24-June-2014

Revision Date 28-April-2022

Revision Number 5

**1. Identification**

**Product Name** Zinc acetate dihydrate

**Cat No. :** AC207640000; AC207640010; AC207640050; AC207645000

**CAS-No** 5970-45-6  
**Synonyms** Acetic acid, zinc salt, dihydrate.

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company****Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number**

For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

**2. Hazard(s) identification****Classification****WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

**Acute oral toxicity** Category 4  
**Serious Eye Damage/Eye Irritation** Category 1

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Harmful if swallowed

Causes serious eye damage

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Rinse mouth

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component              | CAS-No    | Weight % |
|------------------------|-----------|----------|
| Zinc acetate dihydrate | 5970-45-6 | <= 100   |
| Zinc acetate           | 557-34-6  | -        |

### 4. First-aid measures

|                                        |                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms/effects</b> | Causes severe eye damage.                                                                                         |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                       |                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                              |
| <b>Flash Point</b>                    | No information available                                                              |
| <b>Method -</b>                       | No information available                                                              |

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Zinc.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                  |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.                                                                |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.                                                     |

## 7. Handling and storage

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents.                                                           |

## 8. Exposure controls / personal protection

|                            |                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**

|                        |                   |
|------------------------|-------------------|
| <b>Eye Protection</b>  | Goggles           |
| <b>Hand Protection</b> | Protective gloves |

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Natural rubber | See manufacturers recommendations |                 | Splash protection only |
| Nitrile rubber |                                   |                 |                        |
| Neoprene       |                                   |                 |                        |

PVC

-

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

## 9. Physical and chemical properties

|                                        |                                    |
|----------------------------------------|------------------------------------|
| Physical State                         | Solid                              |
| Appearance                             | White                              |
| Odor                                   | vinegar-like                       |
| Odor Threshold                         | No information available           |
| pH                                     | 6.0-7.0 (@ 25 ) 5% in water (25°C) |
| Melting Point/Range                    | 237 °C / 458.6 °F                  |
| Boiling Point/Range                    | No information available           |
| Flash Point                            | No information available           |
| Evaporation Rate                       | Not applicable                     |
| Flammability (solid,gas)               | No information available           |
| Flammability or explosive limits       |                                    |
| Upper                                  | No data available                  |
| Lower                                  | No data available                  |
| Vapor Pressure                         | No information available           |
| Vapor Density                          | Not applicable                     |
| Specific Gravity                       | 1.840                              |
| Solubility                             | Soluble in water                   |
| Partition coefficient; n-octanol/water | No data available                  |
| Autoignition Temperature               | No information available           |
| Decomposition Temperature              | No information available           |
| Viscosity                              | Not applicable                     |
| Molecular Formula                      | C4 H6 O4 Zn . 2 H2 O               |
| Molecular Weight                       | 219.5                              |

## 10. Stability and reactivity

|                                  |                                                               |
|----------------------------------|---------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                    |
| Stability                        | Stable under recommended storage conditions.                  |
| Conditions to Avoid              | Incompatible products. Excess heat. Avoid dust formation.     |
| Incompatible Materials           | Strong oxidizing agents                                       |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Zinc |

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component              | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|------------------------|--------------------------|-------------|-----------------|
| Zinc acetate dihydrate | LD50 = 794 mg/kg ( Rat ) | Not listed  | Not listed      |
| Zinc acetate           | LD50 = 663 mg/kg (Rat)   | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Irritating to eyes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component              | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------------|-----------|------------|------------|------------|------------|------------|
| Zinc acetate dihydrate | 5970-45-6 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Zinc acetate           | 557-34-6  | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component              | Freshwater Algae | Freshwater Fish                                                                              | Microtox   | Water Flea |
|------------------------|------------------|----------------------------------------------------------------------------------------------|------------|------------|
| Zinc acetate dihydrate | Not listed       | LC50: 0.88 mg/l/96 H<br>(Pimephales proelas)<br>LC50: 0.55mg/l/96 H<br>(Onchorynchus mykiss) | Not listed | Not listed |

**Persistence and Degradability** May persist based on information available.

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                             |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its water solubility. |

### 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 14. Transport information

#### DOT

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>UN-No</b>                | UN3077                                              |
| <b>Proper Shipping Name</b> | Environmentally hazardous substances, solid, n.o.s. |
| <b>Technical Name</b>       | (ZINC ACETATE DIHYDRATE)                            |
| <b>Hazard Class</b>         | 9                                                   |
| <b>Packing Group</b>        | III                                                 |

#### TDG

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>UN-No</b>                | UN3077                                              |
| <b>Proper Shipping Name</b> | Environmentally hazardous substances, solid, n.o.s. |
| <b>Hazard Class</b>         | 9                                                   |
| <b>Packing Group</b>        | III                                                 |

#### IATA

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>UN-No</b>                | UN3077                                              |
| <b>Proper Shipping Name</b> | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.* |
| <b>Hazard Class</b>         | 9                                                   |
| <b>Packing Group</b>        | III                                                 |

#### IMDG/IMO

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>UN-No</b>                | UN3077                                              |
| <b>Proper Shipping Name</b> | Environmentally hazardous substances, solid, n.o.s. |
| <b>Hazard Class</b>         | 9                                                   |
| <b>Packing Group</b>        | III                                                 |

### 15. Regulatory information

#### International Inventories

| Component              | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Zinc acetate dihydrate | 5970-45-6 | X   | -    | -    | -                                             | -         | -      | -   |
| Zinc acetate           | 557-34-6  | X   | -    | X    | ACTIVE                                        | 209-170-2 | -      | -   |

| Component              | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Zinc acetate dihydrate | 5970-45-6 | X     | -        | -    | -    | X    | X    | X     | X     |
| Zinc acetate           | 557-34-6  | X     | KE-35519 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous

Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component              | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Zinc acetate dihydrate | Part 1, Group A Substance                            |                                                                            |                                           |
| Zinc acetate           | Part 1, Group A Substance                            |                                                                            |                                           |

**Legend** NPRI - National Pollutant Release Inventory

#### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component              | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Zinc acetate dihydrate | 5970-45-6 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Zinc acetate           | 557-34-6  | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component              | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Zinc acetate dihydrate | 5970-45-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y23                      |
| Zinc acetate           | 557-34-6  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y23                      |

## 16. Other information

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#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**